

CorpusSLR: from one structured query to a PRISMA-compliant, deduplicated corpus

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Abstract

A systematic literature review rests on a multi-database search, but the databases disagree about query syntax, field semantics, paging and export format, and every disagreement is paid for in manual work or in silently lost records. CorpusSLR is a Python library that compiles one structured query into the syntax of each supported database, retrieves through native APIs or parses the file exports of databases without an open API, deduplicates the result with a cascading and fully auditable procedure, and generates the PRISMA 2020 flow diagram and the PRISMA-S search appendix from recorded provenance rather than reconstructing them afterwards. Deduplication reaches F1 0.9996 (0 false positives, 1 false negative) on the published ASySD Diabetes gold standard, and a median pairwise F1 of 0.9592 across fifteen disciplines (13 809 live records, 2584 judgeable pairs) under an identifier-blind protocol. The merged corpus is written in Scopus export layout, verified by executing `bibliometrix::convert2df()` on it. The package depends only on the standard library plus `requests` and runs on Python 3.9 and later.

Keywords: systematic literature review, PRISMA, PRISMA-S, deduplication, bibliographic databases, evidence synthesis

1. Motivation and significance

A systematic literature review stands or falls on its search, in medicine and equally in fields that adopted the method later, such as software engineering [1]. Two decades of methodological work have established that a search resting on a single bibliographic database does not achieve acceptable recall. In a sample of 58 published reviews covering 1746 relevant references, Bramer et al. [2] found that adequate recall required the *combination* of Embase, MEDLINE, Web of Science Core Collection and Google Scholar, reaching 98.3% overall recall; 16% of included references (291 of 1746) were found in one database only, and roughly 60% of published reviews fail to retrieve 95% of the available relevant literature, mostly through omitted databases.

Automation is already used by review teams [3], but the established tools address the stage after retrieval: screening applications such as Rayyan [4] and libraries such as revtools [5] take a corpus as given, and review workbenches administer protocol and screening around a corpus already assembled [6, 7]. Multi-database searching is therefore a methodological requirement rather than a convenience, and it is where the practical work lies. The databases disagree about query syntax, field semantics, paging and export format, and a review team pays for each disagreement either in manual effort or in records that disappear without a trace. The systems also differ in what they cover and how selectively: Singh et al. [8] report that 99.11% of the journals indexed in Web of Science are also indexed in Scopus, so the two are far from interchangeable and the containment is markedly asymmetric; a comparison of five sources at scale reports the same kind of disagreement [9].

Reporting requirements have tightened in parallel. PRISMA 2020 [10] requires a flow diagram accounting for every record from identification to inclusion, and the search extension PRISMA-S [11] requires each database with its platform and interface, the date run, the strategy as executed, the per-database counts and the deduplication method. Assembling that appendix by hand afterwards from screening spreadsheets is error-prone and a common source of reviewer objections.

CorpusSLR addresses the whole chain in one dependency-light library (Table 1): one structured query object compiles to each database's own syntax, records arrive through native APIs or through parsers for the file exports of databases without an open API, deduplication is cascading and fully auditable, and the PRISMA 2020 diagram and PRISMA-S appendix are generated from recorded provenance.

2. Software description

2.1. Architecture

A `SearchQuery` holds blocks of synonyms, a year range, document types and languages, and compiles to Scopus, PubMed, OpenAlex, Crossref, Semantic Scholar, arXiv and Web of Science syntax. Anything it cannot express faithfully in a given dialect produces a warning that is carried into the PRISMA-S appendix, so that a compilation compromise is reported rather than hidden.

A `Corpus` records every search event: database, platform, interface, query as executed, date and hit count, which are exactly the PRISMA-S items. Records are normalised `Record` data-classes. DOIs are lowercased, percent-decoded and stripped of resolver prefixes; titles are transliterated before Unicode decomposition, so that stroked and ligature letters survive normalisation instead of collapsing into gaps.

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Table 1: Code metadata.

Nr	Code metadata description	Please fill in this column
C1	Current code version	1.0.0
C2	Permanent link to code/repository used for this code version	https://github.com/s-matysik/CorpusSLR
C3	Permanent link to Reproducible Capsule	notebooks/corpussl_r_colab.ipynb in the repository, a 26-cell Google Colab notebook running the whole workflow
C4	Legal Code License	MIT License
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python (≥ 3.9)
C7	Compilation requirements, operating environments and dependencies	Pure Python, no compilation. Runtime dependency: <code>requests</code> ≥ 2.28 . Optional: <code>python-docx</code> ≥ 1.1 for the <code>.docx</code> appendix. Development: <code>pytest</code> , <code>pytest-cov</code> . OS independent
C8	If available, link to developer documentation/manual	<code>docs/user_guide.md</code> and <code>docs/api_reference.md</code> in the repository
C9	Support email for questions	Via the repository issue tracker

2.2. Retrieval and parsing

Ten API client classes cover the systems reachable programmatically, nine of which are selectable as databases from the command line. Databases without an open API are supported through their file exports, which is where most of the engineering effort sits: encodings, dialects and vendor quirks that quietly drop records. Coverage of the target specification is 32 of 32 items, itemised in `validation/parser_coverage.csv`.

Import accounting is treated as a correctness property. Every parser maintains the invariant that inputs read equals records returned plus rejections counted with a reason, so a number entering a PRISMA diagram can be reconciled against the file it came from. Four silent-loss defects were found this way: a UTF-16 export yielding zero records, a truncated tagged export losing the whole file, one oversized field aborting a table, and a preprint client stopping after the first page.

The source registry classifies each of its 21 entries as principal or supplementary following Gusenbauer and Hadaway [12], who investigated 28 academic search systems for systematic search qualities, and records *how* each classification is evidenced: named in their principal list, inherited from an access platform they assessed, or, for IEEE Xplore, following discipline-specific guidance that their assessment does not support. A strategy audit warns when a corpus rests on no principal system, on exactly one, or is dominated by supplementary records.

2.3. Deduplication: what is standard and what is not

The core is textbook and is not claimed as a contribution: DOI normalization, title-token blocking, Ratcliff-Obershelp similarity, and union-find for transitive closure. Multi-round blocking is taken from ASySD [13] and credited as such in the source; blocking itself is a general entity-resolution device, available as a standalone package [14]. The problem is under active attention elsewhere [15].

What is new is *negative evidence*: typed, logged separators that refuse a merge the positive evidence would otherwise allow. The contrast with ASySD is specific rather than rhetorical. Read from its source, its rule decides a pair by a disjunction

of 23 clauses over per-field Jaro-Winkler similarities, and it already carries two separators of its own, a mismatching DOI and a year gap greater than one; its DOI clause also requires author and title agreement, so a shared identifier is not decisive there either. What it does not model is the bibliographic locus, page fields mangled by a spreadsheet round trip, pseudo page ranges, a conference version against a journal version, or the instalment markers of a recurring publication. Each separator below was derived from a failure observed in a real corpus rather than postulated.

1. **A missing-value literal is not an identifier.** Exports write “no DOI” as a literal: R writes NA, others N/A, NULL or -. Accepting those as identifiers makes every such record a duplicate of every other. In the ASySD Diabetes file, 492 of 1845 records (26.7%) carry such a literal in the DOI field.
2. **A shared identifier is refused when the records sit at different places in the volume** (the locus guard). One conference-supplement DOI can span an entire session of distinct abstracts.
3. **Conflicting identifiers block a merge**, so that Part 1 and Part 2 of a paper are not fused when a fuzzy title match would otherwise join them.
4. **A title that is only an identifier does not drive similarity.** Some publishers deposit the DOI into the title field; two such strings from one issue differ only in their last digits.
5. **Instalments of a recurring publication are separated by the marker naming the instalment**, which is often the only thing distinguishing consecutive quarters of a report series.

Transcribing that rule set and asking both about identical candidate pairs (`validation/compare_asysd_rule.py`) quantifies it across the fifteen disciplines: median pairwise F1 0.9164 against 0.9510, ours higher in 13 of 15. Neither dominates; the operating points differ, median recall 0.861 against 0.964 at median precision 0.974 against 0.962. Two caveats bound that. Only the automatic decision is compared, and 424 of the 473 pairs the published rule leaves behind (89.6%) fall

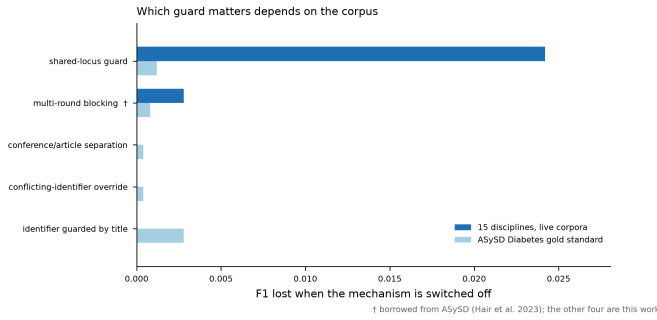


Figure 1: Contribution of each deduplication mechanism, measured by switching it off and rescore. Bars are F1 lost. Multi-round blocking is borrowed from ASySD; the other four are contributed here. The ranking inverts between the two corpora, which is the argument for validating on more than one benchmark.

in the set it routes to manual review, 2974 pairs for a human to inspect here. Its scoring of a field absent on both sides as agreement admits 68 of its 78 false merges, but forcing those to no-evidence would cost 575 true pairs, so that policy buys recall rather than being a defect. On the biomedical gold standard the two sit within 0.0006 F1, so this is about where each rule was tuned, not a ranking.

Each separator’s contribution is measured by disabling it and rescore (Table 2, Figure 1), and the measurement carries a finding: *which one matters depends on the corpus*. The title-guarded identifier match dominates on the gold standard (0.0028 F1 lost when removed) and the locus guard on the fifteen disciplines (0.0242 against 0.0012), so a package validated on one benchmark would have shipped the wrong defaults.

2.4. Reporting and reproducibility

Outputs are the PRISMA 2020 flow diagram (SVG), the PRISMA-S appendix (Markdown or .docx), a per-source metadata completeness report, an empirical cross-source overlap matrix, the full deduplication decision log, and exports to CSV, screening CSV, RIS and BibTeX. A harvest layer archives raw API responses with a manifest and an order-independent checksum, so that a reviewer can replay a search and verify that the corpus is the one the review describes.

2.5. The canonical output: Scopus CSV

A review does not end at a deduplicated corpus; it continues into screening and bibliometric analysis, and those tools read one dialect. CorpusSLR therefore writes the merged corpus in Scopus export layout: 46 columns in the vendor order, which is the input format that bibliometrix, VOSviewer and Embed-SLR [16, 17] accept. The last is the immediate reuse path: it screens a Scopus-layout corpus by sentence embeddings, so this library’s output is its input unconverted, and bibliographic analysis of the same file is served by packages such as Tech-Miner [18].

The claim is verified by execution rather than by a column list. On a live three-database corpus written by `to_scopus_csv()`, `bibliometrix::convert2df(dbsource="scopus")`

under R 4.5.3 imported 166 rows with no record dropped and generated 57 bibliometrix fields; `biblioAnalysis()` then ran on the result, and the institutional collaboration network came out as 156 by 156 institutions. Doing this found a defect that a column-presence check cannot see: the Record dataclass had no affiliations field, so the C1 column was always empty and the institutional network returned an empty matrix. An affiliations field was added, populated from the Web of Science Expanded addresses node, unioned on merge and mapped to the Affiliations column.

Columns with no counterpart in the harvested metadata, namely index keywords and reference lists, are written empty rather than invented. No API in the package returns Scopus index terms or reference lists. An empty column is missing data the analyst can see; a fabricated affiliation or reference list is a false finding inside someone else’s co-citation network.

2.6. Two interfaces for researchers who do not write Python

The command `python -m corpusslr` opens a guided terminal menu built on the standard library alone: no curses, no third-party console library, no ANSI colour, 80 columns. It walks the user through database selection, query construction, API keys (read with `getpass`, stored with mode 0600, never echoed), a dry-run plan, retrieval and export. A Google Colab notebook covers the same workflow in 26 cells for users without a local Python installation.

3. Illustrative examples

3.1. A minimal review in Python

Listing 1 is the library path end to end: compile a query, record two results with their PRISMA-S provenance, deduplicate, and write the Scopus CSV, the flow diagram and the search appendix. The two records are supplied directly so that the example runs offline, without an API key; in a review they would come from a client such as `ScopusSource(api_key=...).search(query)`. The example is kept runnable as `paper/listing_example.py` in the repository.

```
from corpusslr import (Corpus, PrismaFlow, Record, SearchQuery,
                      deduplicate, prisma_s_markdown, to_scopus_csv)

query = SearchQuery(
    blocks=[["deep learning", "convolutional neural network"],
            ["diabetic retinopathy"]],
    years=(2020, 2024), doc_types=["article"],
)
print(query.to_scopus())

corpus = Corpus()
corpus.add_records(
    [Record(title="Deep learning for diabetic retinopathy",
            doi="10.1000/ABC.123", year=2021, source="Scopus")],
    database="Scopus", platform="Elsevier",
    interface="Scopus Search API",
    query=query.to_scopus(), date_run="2026-09-07")
corpus.add_records(
    [Record(title="Deep learning for diabetic retinopathy",
            doi="https://doi.org/10.1000/abc.123", year=2021,
            source="PubMed")],
    database="PubMed", platform="NCBI", interface="E-utilities",
    query=query.to_pubmed(), date_run="2026-09-07")

result = deduplicate(corpus)
print(corpus.total_identified(), "identified ->",
```

Table 2: Contribution of each deduplication mechanism, measured by switching it off and rescoring. Values are F1 lost relative to the shipped defaults. The mechanism marked † is borrowed from ASySD; the other four are contributed here. Data: `validation/dedup_mechanism_ablation.csv` and `validation/dedup_ablation_domains.csv`.

Mechanism	Origin	Δ F1 gold	Δ F1 15 dom.
identifier guarded by title	this work	0.0028	0.0
shared-locus guard	this work	0.0012	0.0242
multi-round blocking †	ASySD	0.0008	0.0028
conflicting-identifier override	this work	0.0004	0.0
conference/article separation	this work	0.0004	0.0

```
len(result.records), "unique")
print("merged by:", result.report.by_method)

to_scopus_csv(result.records, "corpus_scopus.csv")
flow = PrismaFlow.from_dedup(corpus, result)
flow.to_svg("prisma2020_flow.svg")
open("prisma_s.md", "w").write(prisma_s_markdown(corpus, result))
```

Listing 1: A complete review in the library API. Runs offline; the file is executed as part of preparing this article.

Its output shows the compiled Scopus query and the merge:

```
(TITLE-ABS-KEY("deep learning" OR "convolutional neural network"))
AND TITLE-ABS-KEY("diabetic retinopathy")
AND (PUBYEAR > 2019 AND PUBYEAR < 2025) AND (DOCTYPE(ar))
2 identified -> 1 unique
merged by: {'doi': 1}
```

Listing 2: Output of Listing 1, as executed.

The two DOIs differ in case and in the resolver prefix, which is exactly the normalisation the identifier stage has to get right before any similarity is computed.

The same review runs from the command line against a JSON configuration. The check subcommand compiles the per-database queries and prints a configuration checksum without issuing a single request, which is the artefact PRISMA-S Item 8 asks for:

```
$ corpusslr check -c review.json
configuration review.json is valid
warning: no principal search system among the API databases
(Scopus, Web of Science); Gusenbauer & Haddaway (2020)
recommend at least one principal source
{
  "config": {
    "config_sha256": "c4240dae9b6f5a3b896accda3a350cc...",
    "corpusslr_version": "1.0.0",
    "run_at": "2026-09-08T00:43:10Z"
  },
  "databases": [
    {"database": "Crossref", "keyless": true,
     "filters": "from-pub-date:2020-01-01,type:journal-article"},
    ...
  ]
}
```

Listing 3: Dry-run plan from the command-line interface.

The registry warning in that output is the methodological guardrail of Section 2 firing at the point where the strategy can still be changed, rather than in peer review.

3.2. Deduplication against a published gold standard

The guards of Section 2.3 were evaluated on the labelled *Diabetes* set of Hair et al. [13] (1845 records, 1261 duplicates, 584 unique works), the benchmark that study uses for ASySD,

Table 3: Deduplication accuracy on the ASySD *Diabetes* gold standard ($N = 1845$). Rows for the other tools are as published by Hair et al. [13] on the same data; their recall is recomputed from the published confusion matrices, because the source tabulates precision and F1 only, and rounding in the source makes those recomputed values differ marginally from the published ones. Data: `validation/asysd_metrics.csv`, `validation/na_doi_fix_impact.csv`.

Method	FP	FN	Prec.	Rec.	F1
CorpusSLR 1.0	0	1	1.0000	0.9992	0.9996
ASySD	0	2	1.000	0.9984	0.999
EndNote	0	43	1.000	0.9659	0.983
SRA-DM	70	114	0.942	0.9096	0.926
Human reviewers	3	368	0.997	0.7082	0.828

EndNote [19], SRA-DM [20] and human reviewers. Table 3 reports the result.

CorpusSLR matches the published precision of the best tool on this benchmark, with no false positives and one missed duplicate against ASySD’s two. Reaching that was not threshold tuning. Rejecting the literal NA as an identifier removed 492 spurious links in this file alone, moving the measurement from one false positive and nine misses to none and four (`validation/na_doi_fix_impact.csv`); the separators described above account for the rest. Two of the five error classes fixed to reach this level were not threshold problems but normalisation defects that no amount of tuning would have removed: percent-encoded DOIs creating a second identity for the same work, and page fields carrying an article length rather than a location.

3.3. Transfer across fifteen disciplines

Accuracy measured on one biomedical benchmark says nothing about a review in history or logistics, and no labelled duplicate set exists for most fields. The protocol therefore builds ground truth from the one arbiter every database agrees on, the normalized DOI, and then *deletes every identifier field*, so that deduplication must reconstruct the clusters from title, authors, year and bibliographic coordinates alone.

Fifteen disciplines were harvested live from four databases, two queries each: 13 809 records and 2584 judgeable pairs. Median pairwise F1 is 0.9592 and every discipline scores above 0.85 (Table 4, Figure 2).

What the errors are matters more than the aggregate. Of 154 false merges, 141 are version variants of one study, such as a preprint and its published article, two revisions of one deposit,

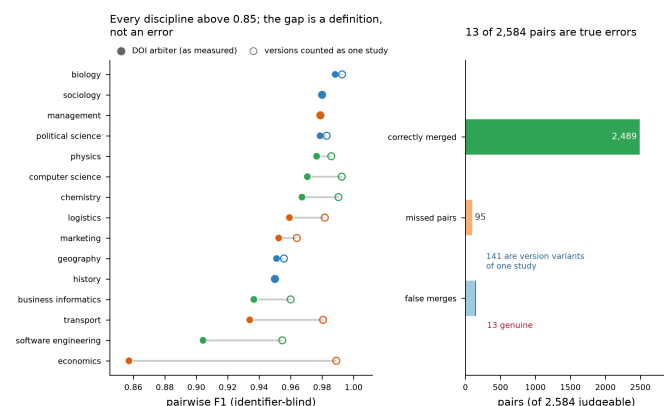


Figure 2: Deduplication across fifteen disciplines, evaluated identifier-blind on 13 809 live records from four databases. Left: pairwise F1 per discipline; the filled marker scores against the DOI arbiter, the open marker treats versions of one study as one study, and the gap between them is a definition rather than an error. Right: composition of the 2584 judgeable pairs; of 154 false merges, 141 are version variants and 13 are genuine over-merges. Colour marks the harvesting track, not a performance grouping.

or two editions of a chapter, and only 13 are genuine over-merges, which is 0.50% of the judgeable pairs. Cochrane treats versions of one study as one study, so counting them as errors measures the arbiter rather than the implementation. Both figures are reported, in the last two columns of Table 4, because the choice belongs to the review team; the median under the version-tolerant definition is 0.9806.

The variation is almost entirely in precision (0.7627 to 1.0000) rather than recall (0.9195 to 1.0000), and it tracks publishing culture rather than subject matter: economics scores lowest because its literature circulates as working papers before publication. Coverage of the DOI arbiter does not predict accuracy, since it limits how much of a corpus can be *judged*, not how well it is deduplicated: computer science scores F1 0.9706 and history 0.9500.

3.4. Scale and live behaviour

On synthetic corpora with 20% planted cross-source duplicates, deduplication processes 60 000 records in 25.3 s at 310 MB peak (422.3 microseconds per record), while a pathological corpus in which every title shares a stock opening takes 9.6 s and 109 MB. Unit cost rises from 260.3 to 422.3 microseconds per record over a tenfold increase in corpus size, so scaling is close to linear but not exactly so. The cost is dominated by title comparison, and memoising the normalized title is what makes it tractable.

Running the retrieval layer against the live services exposed three defects that offline tests cannot reach: the Scopus client defaulted to the leaner response view, which returns one author and no abstract at all, making title and abstract screening impossible; the preprint client had no scan budget, so a selective query over a multi-year window became an hours-long full-archive crawl; and the Web of Science client addressed the Starter API while an entitled subscription serves Expanded. On a live Web of Science Expanded query, the client returned an abstract on 100 of 100 records where Starter returns none.

4. Impact

The measurements of Section 3 support three claims of impact. The immediate one is on compliance work that currently consumes reviewer and author time. The PRISMA-S appendix is generated from recorded provenance, so the database, platform and interface triple, the executed strategy, the dates and the per-database counts cannot drift from what was actually run. Deduplication performs at the level of the best published tool while producing an auditable decision log, which converts “duplicates were removed in EndNote” into a reportable and checkable method.

The second impact is methodological guardrails. The registry makes the principal and supplementary distinction operational rather than advisory: a corpus built only from supplementary systems raises a critical warning at the point where it can still be fixed. To our knowledge this classification has not been automated in a review-support library before.

The third is reproducibility, in the sense of the FAIR principles [21], at two levels. A harvest archive with a manifest and an order-independent checksum lets a reviewer replay a search and confirm the corpus. And a whole review, from search through parsing, deduplication, diagram and appendix, runs from one JSON configuration through the command-line interface, so that the reported method is the method that executed. The dry-run mode of Listing 3 compiles the per-database queries with a configuration checksum before any request is made.

Because the canonical export is the Scopus layout, the corpus continues directly into the established bibliometric toolchain rather than terminating in a bespoke format, which is what makes the package usable as one stage of an existing workflow instead of a replacement for it.

5. Conclusions

CorpusSLR covers the span from a structured query to a PRISMA-compliant, deduplicated and auditable corpus, which is the chain Section 1 identifies as unautomated, and it delivers the reporting artefacts of Section 4 from recorded provenance. Deduplication is validated against a published gold standard, where it reaches F1 0.9996, and across fifteen disciplines under an identifier-blind protocol, where the median pairwise F1 is 0.9592 and the 154 false merges resolve into 141 version variants and 13 genuine over-merges.

The limitations are stated plainly. The cross-disciplinary arms are small in pairs judged per discipline, between 66 and 344, and no confidence intervals are reported. Citation-based strategies such as snowballing [22] are outside its scope. The-saurus expansion, such as MeSH or Emtree, is not performed, so a librarian-designed strategy still outperforms a compiled one in biomedical databases. The measured accuracy across disciplines depends on DOI coverage in the harvested sample, which bounds how much of a corpus can be judged at all. And the ablation shows that the ranking of the deduplication guards inverts between the two benchmarks, so the shipped defaults are a compromise rather than an optimum for any single field.

Table 4: Identifier-blind deduplication across fifteen disciplines, on 13 809 records harvested live from four databases. Pairs are judged against the normalized DOI after every identifier field has been removed. $F1_v$ is the version-tolerant variant, which treats versions of one study as one study. Totals: 13 809 records, 2584 pairs; medians $F1$ 0.9592 and $F1_v$ 0.9806. Data: `validation/domains_15_final.csv`.

Discipline	Track	N	Pairs	Prec.	Rec.	F1	$F1_v$
Economics	social	831	230	0.7627	0.9783	0.8572	0.9890
Logistics	social	893	195	0.9543	0.9641	0.9592	0.9817
Management	social	715	242	1.0000	0.9587	0.9789	0.9789
Marketing	social	895	171	0.9697	0.9357	0.9524	0.9639
Transport	social	891	182	0.8985	0.9725	0.9340	0.9806
Business inf.	stem	1053	102	0.9320	0.9412	0.9366	0.9600
Chemistry	stem	914	105	0.9537	0.9810	0.9672	0.9904
Computer sci.	stem	1187	66	0.9429	1.0000	0.9706	0.9925
Physics	stem	1193	319	0.9750	0.9781	0.9765	0.9858
Software eng.	stem	617	149	0.8896	0.9195	0.9043	0.9547
Biology	nat./hum.	1191	344	0.9912	0.9855	0.9883	0.9927
Geography	nat./hum.	889	105	0.9798	0.9238	0.9510	0.9557
History	nat./hum.	819	102	0.9694	0.9314	0.9500	0.9500
Political sci.	nat./hum.	841	119	0.9914	0.9664	0.9787	0.9829
Sociology	nat./hum.	880	153	1.0000	0.9608	0.9800	0.9800

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The evaluation rests on the labelled deduplication benchmark published by the ASySD authors, whose decision to release it openly is what made an external comparison possible. Clarivate granted the Web of Science Expanded API entitlement used to verify that client against the live service.

Declaration of competing interest

The author declares no competing interests.

Data availability

All data underlying the results are public. The software, the evaluation scripts and the measurement tables are in the repository given in C2. The validation data are additionally packaged for review as `corpusslr_validation_data.zip`, whose manifest maps each file to the table or figure it supports. One file is third-party: the ASySD *Diabetes* gold standard, redistributed unmodified from `github.com/camaradesuk/ASySD` under that project’s GPL-3.0 licence, with provenance, pinned commit and checksum recorded in `validation/THIRD-PARTY_DATA.md`. The corpora harvested for the cross-disciplinary study are archived as compressed record dumps, so every number reported here recomputes without network access.

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